

TRAVELGUARD AI

Real-time fraud decisions for traveling bank customers

An AI decision service that combines transaction data with telecom network signals to approve, challenge, or block card transactions in milliseconds.

THE PROBLEM

Banks can't tell the difference between a traveler and a thief

False declines

A customer travels abroad, makes a normal purchase, and the card gets blocked — hurting trust and revenue.

No real-world context

Traditional risk engines only see transaction data — amount, merchant, card — never where the customer's phone actually is.

Missed fraud

A stolen card or SIM-swapped phone is used far from home, and rule-based systems approve it anyway.

Slow, manual reviews

Ambiguous cases go to human review queues, adding delay and cost instead of resolving instantly.

THE SOLUTION

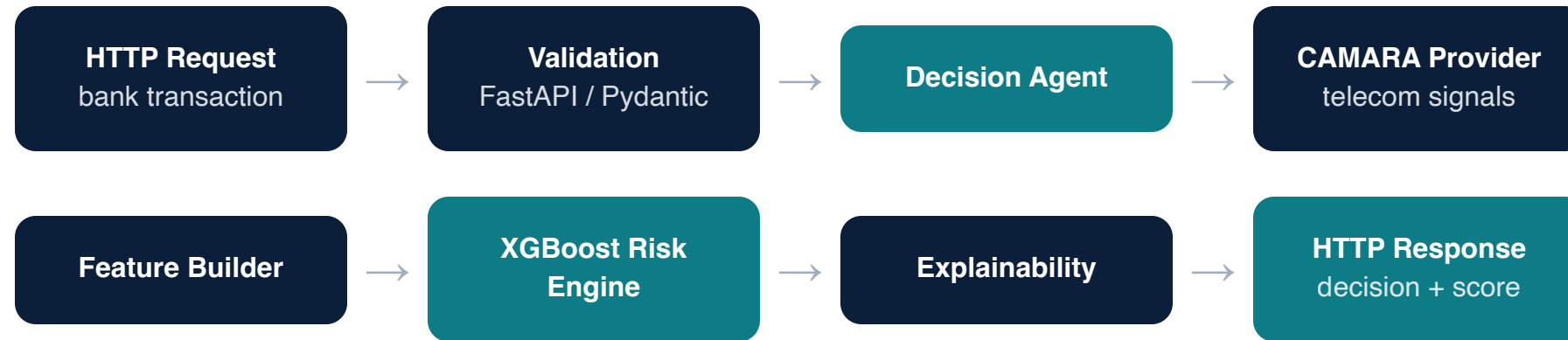
One API call. One decision.

TravelGuard AI wraps a transaction risk engine with live telecom signals — roaming status, device location, SIM-swap history — so every transaction gets a fast, explainable decision instead of a blind guess.

- Single **POST /v1/decision** endpoint for bank-facing systems
- Combines transaction + customer profile + telecom network signals
- Returns a decision, risk score, confidence, and plain-language reasons
- Built for production: health checks, logging, Docker, sub-100ms latency

HOW IT WORKS

The request flow



Every step is measured end-to-end — the response includes total latency in milliseconds for full observability.

Context a bank alone can't see

TravelGuard AI calls standardized CAMARA telecom APIs to enrich every decision with real-world network signals.

Location

Confirms the customer's network-reported country against the transaction's merchant country.

SIM Swap

Flags a recent SIM swap — a strong indicator of account takeover fraud.

Roaming

Detects whether the customer's phone is currently roaming abroad, matching travel to spending.

Number Verification

Confirms the phone number on file truly belongs to the active SIM.

Device Location

Cross-checks device-level location for finer-grained verification.

Trusted Devices

Checks the transaction device against the customer's known, trusted device list.

From risk score to action

A trained XGBoost model scores the transaction; simple, transparent business rules turn that score into one of three outcomes.

APPROVE

risk score < 0.25

Low risk, or a trusted device with high confidence. Transaction proceeds immediately.

STEP_UP

risk score $0.25 - 0.70$

Medium risk. Ask for additional verification (OTP, biometrics) before proceeding.

REJECT

risk score ≥ 0.70 , or recent SIM swap

High risk or a SIM swap in the last 48 hours. Transaction is blocked.

EXPLAINABILITY

Every decision comes with a reason

No black box. Each response lists the top contributing features and business rules that led to the outcome — ready for audit, compliance, or a customer-facing message.

```
{
  "decision": "STEP_UP",
  "risk_score": 0.4123,
  "confidence": 0.88,
  "explanation": [
    "medium_risk_score",
    "feature:amount_ratio_to_average:6.595",
    "feature:trusted_device:1.000"
  ],
  "signals_used": ["location", "roaming", "device_location", "sim_swap"],
  "latency_ms": 42.7
}
```

Simple to integrate

Endpoint	Purpose
POST /v1/decision	Main decision API — transaction, customer profile, signals in; decision out
GET /health	Liveness check for orchestration / load balancers
GET /ready	Readiness check — model and telecom provider availability
GET /docs	Interactive Swagger UI, auto-generated from the API schema

Clear, typed error codes for every failure mode:

400 INVALID_REQUEST

422 VALIDATION_ERROR

503 MODEL_UNAVAILABLE

503 PROVIDER_UNAVAILABLE

500 INFERENCE_ERROR

BUILT FOR PRODUCTION

Technology stack

API Layer

FastAPI + Pydantic on Python 3.12, with request logging (request id, status, decision, latency) and dependency-injected singletons.

Risk Model

XGBoost risk engine trained on transaction and behavioral features, wrapped by a rule-based decision layer.

Telecom Integration

CAMARA-standard provider abstraction, with support for Open Gateway and Nokia providers plus a mock provider for testing.

Deployment

Dockerized service, ready to run locally or in any container orchestrator; full pytest suite for confidence.

TRAVELGUARD AI

**Faster approvals.
Fewer false declines.
Real fraud caught.**

A single decision service that lets banks say yes to their traveling customers with confidence — backed by telecom-grade signals and explainable AI.

Thank you